MMP 1
Serie 3
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Kannst du mir sagen, ob bei den Beweisen alles Wichtige dabei ist.
Gibt es eine Strategie die Funktion wie in Aufgabe 4 zu erraten?
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0.a

(D)

1

1.a

$$f(x) = \frac{1}{2} - x + \lfloor x \rfloor \tag{1.1}$$

$n \neq 0$

$$f_n = \int_0^1 \left(\frac{1}{2} - x\right) e^{-2i\pi x n} dx = \left[\frac{i}{4\pi n} e^{-2\pi i n x} - \left(\frac{x}{-2\pi i n} e^{-2\pi i n x} - \int_0^1 \frac{1}{-2\pi i n} e^{-2\pi i n x} dx\right)\right]_0^1 = \frac{1}{2\pi i n} \tag{1.2}$$

$n = 0$

$$f_0 = \int_0^1 \frac{1}{2} - x dx = \left[\frac{x}{2} - \frac{x^2}{2}\right]_0^1 = 0 \tag{1.3}$$

1.b

$$\begin{aligned} f_n &= \int_{-\frac{1}{2}}^{\frac{1}{2}} \left(\frac{1}{2} - |x|\right) e^{-i2\pi n(x+\frac{1}{2})} dx \\ &= (-1)^n \int_{-\frac{1}{2}}^{\frac{1}{2}} \left(\frac{1}{2} - |x|\right) e^{-2\pi i n x} dx \end{aligned} \tag{1.4}$$

$n = 0$

$$f_0 = \left[\frac{x}{2}\right]_{-\frac{1}{2}}^{\frac{1}{2}} - 2\left[\frac{x^2}{2}\right]_0^{\frac{1}{2}} = \frac{1}{2} - \frac{1}{4} = \frac{1}{4} \tag{1.5}$$

$n \neq 0$

$$\begin{aligned} f_n &= -(1)^n \left(\left[\frac{1}{-4\pi i n} e^{-2\pi i n x}\right]_{-\frac{1}{2}}^{\frac{1}{2}} + \int_0^{\frac{1}{2}} x e^{-2\pi i n x} dx + \int_0^{\frac{1}{2}} x e^{2\pi i n x} dx\right) \\ &= (-1)^n \left(0 + \underbrace{\left[\frac{x}{-2\pi i n} e^{-2\pi i n x} + \frac{x}{2\pi i n} e^{2\pi i n x}\right]_0^{\frac{1}{2}}}_{=0} - \left(\int_0^{\frac{1}{2}} \frac{1}{-2\pi i n} e^{-2\pi i n x} dx + \int_0^{\frac{1}{2}} \frac{1}{2\pi i n} e^{2\pi i n x} dx\right)\right) \\ &= (-1)^n \left(\left[\frac{1}{4\pi^2 n^2}(e^{-2\pi i n x} + e^{2\pi i n x})\right]_0^{\frac{1}{2}}\right) = (-1)^n \left[\frac{1}{2\pi^2 n^2} \cos(2\pi n x)\right]_0^{\frac{1}{2}} = (-1)^n \frac{1 + (-1)^n}{2\pi^2 n^2} \end{aligned} \tag{1.6}$$

1.c

$$f(x) = x^2 \tag{1.7}$$

$$f_n = \frac{1}{2\pi} \int_0^{2\pi} x^2 e^{-inx} dx \tag{1.8}$$

$n = 0$

$$f_0 = \frac{1}{2\pi} \left[\frac{x^3}{3}\right]_0^{2\pi} = \frac{4\pi^3}{3} \tag{1.9}$$

$n \neq 0$

$$
\begin{aligned}
f_n &= \frac{1}{2\pi}\int_0^{2\pi} x^2 e^{-inx}dx \\
&= \frac{1}{2\pi}\left(\left[\frac{x^2}{-in}e^{-inx}\right]_0^{2\pi} - \int_0^{2\pi}\frac{2x}{-in}e^{-inx}dx\right) \\
&= \frac{1}{2\pi}\left(\frac{i4\pi^2}{n} - \left(\left[\frac{2x}{-n^2}e^{-inx}\right]_0^{2\pi} + \int_0^{2\pi}\frac{2}{n^2}e^{-inx}dx\right)\right) \\
&= \frac{1}{2\pi}\left(\frac{i4\pi^2}{n} + \frac{4\pi}{n^2} + \left[\frac{i}{n^3}e^{-inx}\right]_0^{2\pi}\right) = \frac{i2\pi}{n} + \frac{2}{n^2}
\end{aligned}
\tag{1.10}
$$

1.d

$$
f(x) = \begin{cases} 0 & -\pi < x < 0 \\ \sin(x) & 0 \le x < \pi \end{cases}
\tag{1.11}
$$

$$
f_n = \frac{1}{2\pi}\int_0^{2\pi} f(x)e^{-inx}dx
\tag{1.12}
$$

$n = 0$

$$
f_0 = \frac{1}{2\pi}\int_0^{\pi}\sin(x)dx = -\frac{1}{2\pi}[\cos(x)]_0^{\pi} = \frac{1}{\pi}
\tag{1.13}
$$

$n \neq 0, 1$

$$
\begin{aligned}
f_n &= \frac{1}{2\pi}\int_0^{\pi}\sin(x)e^{-inx}dx \\
&= \frac{1}{2\pi}\left(-\cos(x)e^{-inx} - \int_0^{\pi} in\cos(x)e^{-inx}dx\right) \\
&= \frac{1}{2\pi}\left(-\cos(x)e^{-inx} - in\sin(x)e^{-inx} + n^2\int_0^{\pi}\sin(x)e^{-inx}dx\right) \\
&= \frac{1}{2\pi}\frac{1}{1-n^2}\left[-\cos(x)e^{inx} - in\sin(x)e^{-inx}\right]_0^{\pi} = \frac{1}{2\pi(1-n^2)}(1+(-1)^n) \quad \forall n > 1
\end{aligned}
\tag{1.14}
$$

$n = 1$

$$
\begin{gathered}
f_1 = \frac{1}{2\pi}\int_0^{\pi}\frac{1}{2i}\left(e^{ix(1-1)} - e^{-2ix}\right) = \frac{1}{2\pi}\left(\frac{\pi}{2i} + 0\right) = -\frac{i}{4} \\
f_{-1} = \frac{i}{4}
\end{gathered}
\tag{1.15}
$$

2

2.a

$$
f(x) = \sin^N(x)
\tag{2.1}
$$

$$
\begin{aligned}
f_n &= \frac{1}{2\pi}\int_0^{2\pi}\frac{1}{(2i)^N}\left(e^{ix} - e^{-ix}\right)^N e^{-inx}dx \\
&= \frac{1}{2\pi 2^N i^N}\int_0^{2\pi}\sum_{j=0}^{N}(-1)^j\binom{N}{j}e^{ix(N-2j-n)}dx \\
&= \frac{2\pi}{2\pi 2^N i^N}(-1)^{\frac{N-n}{2}}\binom{N}{\frac{N-n}{2}} \\
&= \frac{1}{2^N i^N}(-1)^{\frac{N-n}{2}}\binom{N}{\frac{N-n}{2}}, \quad \forall\frac{N-n}{2} \in \mathbb{N}
\end{aligned}
\tag{2.2}
$$

2.b

$$
f(x) = e^{e^{ix}} = \sum_{j=0}^{\infty}\frac{e^{ixj}}{j!}
\tag{2.3}
$$

$$f_n = \frac{1}{2\pi}\int_0^{2\pi}\sum_{j=0}^{\infty}\frac{e^{ix(j-n)}}{j!}dx = \frac{1}{n!} \tag{2.4}$$

2.c

$$f(x) = \frac{1}{1-\frac{1}{2}e^{ix}} = \sum_{j=0}^{\infty} 2^{-j}e^{ixj} \tag{2.5}$$

$$f_n = \frac{1}{2\pi}\int_0^{2\pi}\sum_{j=0}^{\infty} 2^{-j}e^{ix(j-n)}dx = 2^{-n} \tag{2.6}$$

2.d

$$f(x) = \frac{4-2\cos(x)}{5-4\cos(x)} = \frac{(5-4\cos(x))\frac{1}{2}+\frac{3}{2}}{5-4\cos(x)} = \frac{1}{2}+\frac{3}{2}\frac{1}{5-4\cos(x)} \tag{2.7}$$

$$5-4\cos(x) = 5-2\big(e^{ix}+e^{-ix}\big) = 4-2\big(e^{ix}+e^{-ix}\big)+1 = \big(2-e^{ix}\big)\big(2-e^{-ix}\big) \tag{2.8}$$

$$\Rightarrow f(x) = \frac{1}{2}+\frac{3}{4}\frac{1}{1-\frac{1}{2}(e^{ix})}\frac{1}{1-\frac{1}{2}e^{-ix}} = \frac{1}{2}+\frac{3}{8}\sum_{j,k\geq 0}\left(\frac{1}{2}\right)^{j+k}e^{ix(j-k)} \tag{2.9}$$

$$\begin{aligned} f_n &= \frac{1}{2\pi}\int_0^{2\pi}\frac{1}{2}e^{-inx}+\frac{3}{8}\sum_{j,k\geq 0}\left(\frac{1}{2}\right)^{j+k}e^{ix(j-k-n)}dx \\ &= \frac{\delta_{0n}}{2}+\frac{3}{8}\sum_{j\geq 0}\left(\frac{1}{2}\right)^{2j+n} \\ &= \frac{\delta_{n0}}{2}+\frac{3}{8}\left(\frac{1}{2}\right)^n\frac{1}{\frac{3}{4}} = \frac{1}{2}(\delta_{0n}+2^{-n}) \end{aligned} \tag{2.10}$$

3

3.a

$$f(x) = \sum_n f_n e^{\frac{2\pi inx}{L}}, \quad \text{with}\sum_n |f_n| < \infty \tag{3.1}$$

$$\begin{aligned} \langle f\ ,\ g\rangle &= \frac{1}{L}\int_0^L \overline{f(x)}g(x)dx \\ &= \frac{1}{L}\int_0^L \sum_{n\in\mathbb{Z}}\overline{f_z}e^{\frac{-2\pi inx}{L}}g(x)dx \\ &= \sum_{n\in\mathbb{Z}}\overline{f_n}\frac{1}{L}\int_0^L g(x)e^{\frac{2\pi inx}{L}}dx \\ &= \sum_{n\in\mathbb{Z}}\overline{f_n}g_n \end{aligned} \tag{3.2}$$

3.b

$$\frac{1}{L}\int_0^L |f(x)|^2dx = \frac{1}{L}\int_0^L \left|\sum_{n\in\mathbb{Z}}e^{\frac{2\pi inx}{L}}\right|^2 dx = \sum_{n\in\mathbb{Z}}|f_n|^2 \tag{3.3}$$

4

4.a

4.a.i

We have to find a L-periodic function such that

$$f_n = \frac{1}{L}\int_0^L f(x)e^{\frac{-2\pi inx}{L}}dx = \left|\frac{1}{n}\right| \tag{4.1}$$

$f(x) = x$

$$f_n = \frac{1}{2\pi}\int_{-\pi}^{\pi} xe^{-inx}dx = \frac{1}{2\pi}\Big[\frac{x}{-in}e^{-inx}\Big]_{-\pi}^{\pi} - \frac{1}{2\pi}\underbrace{\int_{-\pi}^{\pi}\frac{1}{-n^2\pi^2}e^{-inx}dx}_{=0} \tag{4.2}$$
$$= \frac{1}{in}$$

$$\Rightarrow \sum_{n\in\mathbb{N}} = \frac{1}{4\pi}\int_{-\pi}^{\pi} x^2dx = \frac{1}{4\pi}\left[\frac{x^3}{3}\right]_{-\pi}^{\pi} = \frac{\pi^2}{6} \tag{4.3}$$

4.a.ii

$$f(x) = (1-2x)^2 \tag{4.4}$$

$n \neq 0$

$$\begin{aligned} f_n &= \int_0^1 (1-2x)^2 e^{-2\pi inx}dx \\ &= \underbrace{\left[\frac{(1-2x)^2}{-(2\pi in)}e^{-inx2\pi}\right]_0^1}_{=0} - \int_0^1 \frac{2(1-2x)}{in\pi}e^{-xin2\pi}dx \\ &= \left[\frac{1-2x}{i^2n^2\pi^2}\right]_0^1 - \underbrace{\int_0^1 \frac{2}{i^2n^2\pi^2}e^{-2\pi inx}dx}_{=0} \\ &= \frac{2}{n^2\pi^2} \end{aligned} \tag{4.5}$$

$n = 0$

$$f_0 = \int_0^1 (1-2x)^2dx = \frac{1}{3} \tag{4.6}$$

$$\begin{aligned} \Rightarrow \sum_{n=1}^{\infty}\frac{1}{n^4} &= \frac{1}{2}\left(\frac{\pi^2}{2}\right)^2\left(\int_0^1 (1-2x)^4dx - |f_0|^2\right) \\ &= \frac{\pi^4}{8}\cdot\left(\frac{9-5}{45}\right) = \frac{\pi^4}{90} \end{aligned} \tag{4.7}$$

4.b

$$f(x) = \begin{cases} 0 & -\pi < x < 0 \\ \sin(x) & 0 \leq x < \pi \end{cases} \tag{4.8}$$

$$f_0 = \frac{1}{\pi} \tag{4.9}$$

$$f_{1,2} = \pm\frac{i}{4} \tag{4.10}$$

$$f_{n>1} = \frac{1}{2\pi(1-n^2)}(1+(-1)^n) \tag{4.11}$$

$$\begin{aligned} \Rightarrow \sum_{n=1}^{\infty}\frac{1}{(4n^2-1)^2} &= \frac{\pi^2}{2}\left(\frac{1}{2\pi}\int_0^{2\pi}|f(x)|^2dx - \frac{1}{\pi^2} - \frac{1}{8}\right) \\ &= \frac{\pi}{4}\int_0^{\pi}\sin^2(x)dx - \frac{1}{2} - \frac{\pi^2}{16} \\ &= \frac{\pi}{4}\int_0^{\pi}\frac{1-\cos(2x)}{2}dx - \frac{1}{2} - \frac{\pi^2}{16} \\ &= \frac{\pi^2}{16} - \frac{1}{2} \end{aligned} \tag{4.12}$$

5

5.a

$$\begin{aligned}(T_a f)_n &= \frac{1}{L}\int_0^L (T_a f) e^{-\frac{2\pi i n x}{L}} dx \\ &= \frac{1}{L}\int_{-a}^{L-a} f(x) e^{-\frac{2\pi i n (x+a)}{L}} = e^{-\frac{2\pi i n}{L} a} f_n\end{aligned} \tag{5.1}$$

5.b

$k \in \mathbb{Z}$

$$\left(e^{\frac{2\pi i k x}{L}} f\right)_n = \frac{1}{L}\int_0^L f(x) e^{-\frac{2\pi i (n-k) x}{L}} dx = f_{n-k} \tag{5.2}$$

5.c

$$f \in \mathcal{F}_L \tag{5.3}$$

$$\begin{aligned}(f')_n &= \frac{1}{L}\int_0^L f'(x) e^{-\frac{2\pi i n x}{L}} dx \\ &= \frac{1}{L}\underbrace{\left[f(x) e^{-\frac{2\pi i n x}{L}}\right]_0^L}_{(5.3)\ \Rightarrow 0} + \frac{2\pi i n}{L}\frac{1}{L}\int_0^L f(x) e^{-\frac{2\pi i n x}{L}} dx \\ &= \frac{2\pi i}{L} n f_n\end{aligned} \tag{5.4}$$